



Gorkha North Field Report Summary

CRS Shelter and Settlements Team

The CRS Nepal shelter program and the CRS Humanitarian Response Department (HRD) conducted a research visit to Northern Village Development Committees (VDCs - government administrative area) of the Gorkha district. The purpose of the visit was to assess status and quality of the housing reconstructions and, if appropriate, make recommendations to improve safety, health and suitability. CRS Nepal's recovery and reconstruction strategy has a strong focus on ensuring houses are built back safer and that the affected population has the knowledge and capacity to design and build appropriate, safe and acceptable housing solutions.

Each village and VDC visited has its own unique constraints, for example the availability of materials, the level of community skill-sets, and income. The four VDCs visited up in the north (Himalayan Region) are not accessible by road, which creates a very different context for support compared to the southern parts of Gorkha district. Throughout the visited VDC's of Sama, Lho, Prok and Bihi typical houses are two storey, double pitched structures, constructed out of dry-stone with no mortar beds to form a rectangular volume, with gables at each end. Intermediary timber floors are not often tied into the stone walls and so do not provide any benefits as a diaphragm. Foundations are typically around 1' deep of un-stepped dry-stone. Verandahs are added to almost all houses with a timber structure set between stone flank or wing walls. These are structurally separate as the stonework is not keyed to the stonework on the main house. Traditionally an independent timber structure sits inside the stone wings to support the first floor. The roof structure is almost always formed using timber in the round. The under-purlins or rafters bear directly atop the stone walls.

The definition of the home as a single structure has also been challenged, with

each household typically owning more than one Khim (Tibetan for house). As the seasons change families move across valleys or to land at different altitudes or orientation. This is tied closely to the farming traditions and needs.

Approximately 70-80% of houses have been rebuilt with the same details as used before their collapse. There was no evidence that seismic improvement details were being incorporated in reconstructions. This lack of compliance to the Nepal Building Code and seismic guidance available is likely to mean the receipt of any government grant will be severely restricted. The government grant is supplied via a bank account. There are no bank branches in the north, and as such due to the remoteness complications may arise in receiving the grant money in a timely manner.

The report tries to identify the unique issues present for each VDC. Village contexts and their differences need to be carefully considered in any future programming or proposals. Recommendations are split into three distinct groups. The first focuses on shelter retrofit options to make structures safer and increase their resilience to earthquakes; these options are the most applicable as the majority of houses have been rebuilt. The second includes recommendations that explore details to be considered when constructing a new house. A third group identifies other recommendations that deal with issues connected to tourism, livelihoods, access, and land management. These groups are set out in the tables overleaf and the key issues observed highlight opportunities and recommendations for any future programming in the Northern VDC's.

Drawings, photos and studies from the field trip can be found within the CRS report 'Gorkha North Field Report June 2016'.

OBJECTIVES

- Assess building typologies and construction techniques, identify good and bad practices, define recommendations for future designs in the area and evaluate techniques for future construction with indigenous material

- Provide recommendation for housing programming in the north for reconstruction, repair and retrofit options, as well as working modalities in the area in terms of field supervision, training and technical assistance.



New CGI roofs replacing vernacular details



Collapsed gables and verandah wing walls



Timber first floor constructions only found on hotel buildings



Open fires for cooking/ heating

GROUP 1 - RETROFIT OPPORTUNITIES

COST	ISSUE	RECOMMENDATION
1	Roof structure movement creates weaker walls as stones will be forced to move at the top of walls.	Use a top/ wall/ ribbon plate - Timber ladder should be constructed along the top of the dry-stone wall providing opportunity for strong fixings to the roof structure
2	Gable stonework is unsupported and is a common point of collapse	Use lightweight gables - Timber truss structure should replace the stonework above the top plate
3	No bracing has been observed to help improve the structures rigidity	Use bracing - Timber structures require bracing in two directions and should follow the rule of thumb illustrated in diagram on page 54
4	Poor bracing within the roof	Use collar ties between roof rafters or bracing as illustrated on p54
5	Ridge beam poorly supported	Use internal posts to support ridge beam
6	Lintels under sized above window openings	Increase quality and size of timber lintels to support stonework above
7	Internal air pollution	Replace open fires with stove and flue for cooking and heating
8	Stone walls at first floor to verandahs collapsing	Replace stonework with timber
9	Dry stonewalls not airtight and are draughty, increasing heat loss.	Apply mud / clay render to walls to prevent seal voids between stones
10	Traditional stone tiled and timber shingle roofs leak and are not airtight.	Support the vernacular traditional building style by maintaining stone tile or timber shingles where possible, issues of water ingress could be mitigated by the inclusion of a moisture control membrane (black) beneath. Green CGI is preferred over blue or natural if used
11	Water is carried a distance from the community water points back to houses.	Rainwater collection could be promoted and storage containers could be supplied and be positioned beneath gutters for each household.
12	In high winds the roof structure and roof covering are often lifted and damaged by up draught	Rafter wall ties should be used to secure the roof structure to the stonewalls and these can be retrofitted by threading wire through existing dry-stone walls

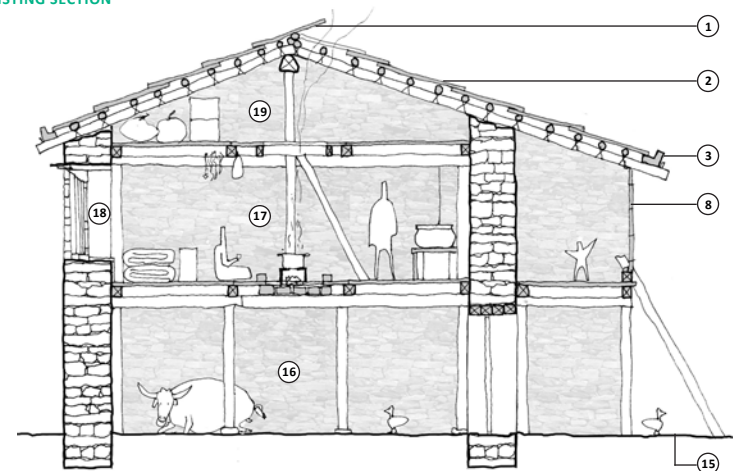
GROUP 2 - NEW HOUSE

COST	ISSUE	RECOMMENDATION
13	Small and poorly shaped stones reduces wall stability and resilience	Use large stones, which are shaped to rectangular forms. Corner stones should extend beyond the wall width, with through stones every 1m2
14	Weak foundations cause instability of entire house	Use stepped foundations at a suitable depth
15	Many new walls lean are not level and lean in places reducing resistance to earthquakes	Use of plumb bob/ line and horizontal levels to set out building elements during construction
16	No timber banding observed	Provide details for timber banding, in areas where timber is scarce steel mesh reinforcement bands set between stone courses could be explored as an alternative
17	Poor stability for floor posts as no connection	Timber posts should be suitably fixed to a stone base pad. Timber posts in a horizontal line should be founded on a timber no smaller than the post size, this detail was not observed. No fixings observed between posts and beams

GROUP 3 - OTHER

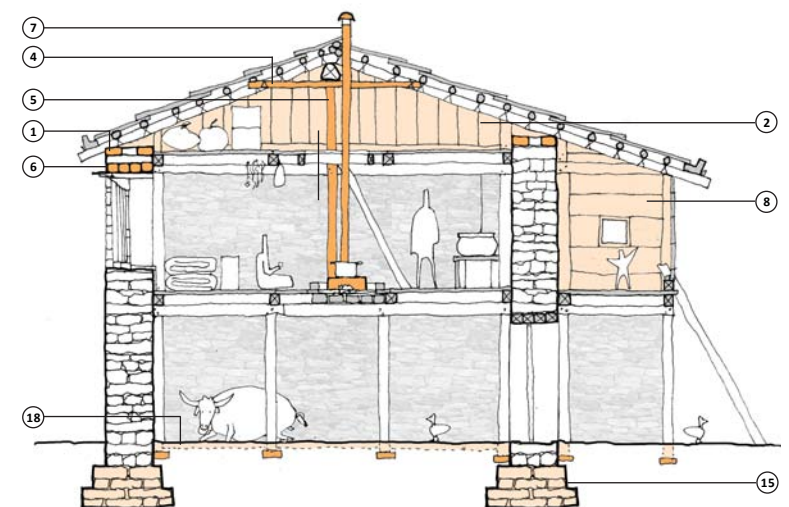
COST	ISSUE	RECOMMENDATION
18	Deforestation, and growing effects from climate change; increased snow melt affecting water table and cleanliness of the water	Support and promote timber use and support forestry programs, community managed and processed
19	Lack of local community seismic construction skills	Support of training programs in carpentry and masonry construction
20	Communities import excessive food items for hotel menus, which are costly and leave a lot of waste packaging	Support the design of menu's that use local produce and recipes
21	Availability of washing facilities	Provide bathing and washing facilities and hygiene information
22	Low availability of fruit	Support fruit tree planting to provide increased opportunity for livelihoods and food security
23	Poor access to information regarding GoN grant and support	Provide access to information services

EXISTING SECTION



- 1 - Roof void chimney
- 2 - Timber planks or stone tile roof covering
- 3 - Timber gutter carved from single trunk
- 4 - Timber verandah
- 5 - Boundary dry-stone wall
- 6 - Stored fire wood
- 7 - Entrance area
- 8 - Outside WC
- 9 - Separate stone wing wall to verandah
- 10 - Projecting stone shelf for tablet carvings
- 11 - Timber ridge beam
- 12 - Timber flag pole lashed to ridge beam
- 13 - Yard space for livestock
- 14 - Timber pegs fixed to support the gutters
- 15 - Compacted earth floor
- 16 - Ground floor livestock space
- 17 - First floor living / sleeping / cooking space
- 18 - Ornamental carved and painted window
- 19 - Attic for storage

REVISED SECTION



NOTES:

The colours down the side try to indicate how costly in time, resources and money certain recommendations are likely to be.

Colour Key:
 Low cost
 Medium cost
 High cost